31 July 1957

MILITARY SPECIFICATION

SWITCH, FIRING, MARK 3 MOD 6 (INERTIA)

(FOR EXPLODER MECHANISM MARK 11 SERIES)

1. SCOPE

1.1 Scope. - This specification states the requirements for the procurement of the Firing Switch Mark 3 Mod 6 , which is an inertia-operated electric switch with normally open contacts, and functions on target impact to complete a firing circuit.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, or such portion thereof as may be designated herein, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-T-18404(NOrd)	Torpedoes, Environmental, Requirements, General Specifications for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC

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DRAWINGS

NAVY DEPARTMENT

BUORD LD 475634	Firing Switch Mark 3 Mod 6 and all parts and subassemblies listed thereon
BUORD DWG 1390702	Firing Switch Assembly
BUORD DWG 1263258	Weight Assembly

PUBLICATIONS

BUREAU OF ORDNANCE

OD 6474	Instructions for Testing Firing Switch Resistance and Sensitivity
OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment
NAVORD OS 6341	Miscellaneous General Design Requirements

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Description. - The Firing Switch Mark 3 Mod 6 shall be an inertia-operated electric switch of the ball type with normally open contacts, and shall consist of a ball weight assembly and contact assembly encased in a hermetically sealed metallic shell. The shell shall be completely encased in a plastic housing. Two wire leads shall be provided, one of them connected internally with the ball weight and the other one grounded to the shell.

3.2 General requirements

3.2.1 Preproduction samples. - Unless otherwise specified in the contract or order, preproduction samples of the firing switch shall be

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manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification. See 4.1.2

3.2.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the firing switch shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUORD LD 475634.

3.2.3 Finishes, coatings, construction and dimensions . - Finishes, coatings, construction and dimensions for the firing switch shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUORD LD 475634.

3.2.4 Moisture, fungus, and corrosion. - The firing switch shall be rendered moisture, fungus and corrosion resistant as required by NAVORD OS 6341 and in accordance with detail requirements specified in the drawings and specifications listed in BUORD LD 475634.

3.2.5 Marking. - Marking of assemblies, subassemblies and parts shall be in accordance with the applicable drawings listed in BUORD LD 475634.

3.2.6 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings will be made without prior revision and correction of the applicable drawings. See 3.2.9

3.2.7 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory

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fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.8 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. - The environmental requirements of this specification are defined in MIL-T-18404(NOrd).

3.4 Performance requirements and product characteristics. - The firing switch shall meet the following performance requirements and product characteristics:

3.4.1 Ball staking. - After the ball is staked in position during the assembly of the weight base, the ball shall resist without perceptible rotation a 2.5 inch pound torque applied in either direction about the axis of the shaft.

3.4.2 Pigtail wire securement. - The pigtail wire, as secured in the weight base by the bushing, shall withstand without loosening or separating from the base a 1 pound pull applied in the direction along the axis of the weight base.

3.4.3 Airtightness. - The firing switch subassembly, BUORD Drawing 1390702 shall evidence no leakage of air when the external air pressure is reduced to 15 inches of mercury.

3.4.4 Sensitivity. - When the firing switch is accelerated in such a direction that the ball is urged along the main axis toward the nose cap, and when it is accelerated in a direction perpendicular to the main axis, the switch shall not close with an applied acceleration up to 3.0 gravity units, shall close instantaneously before the applied acceleration reaches 6.0 gravity units, and shall reach a state of steady closure before the applied acceleration reaches 6.5 gravity units. Instantaneous closure shall be considered obtained when a pip is observed on the oscilloscope during testing as specified in 4.5.3.4.4. Steady closure shall be considered obtained when a straight

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line (with possible hash due to equipment circuitry) is observed on the same oscilloscope screen.

3.4.5 Contact resistance. - The contact resistance of the switch during steady closure shall be not more than 0.75 ohms as measured between the two wire leads from the switch.

3.4.6 Dielectric strength. - When the firing switch is in the normally open condition, the insulation resistance, as measured between the two wire leads of the switch, shall be not less than 10 megohms under an applied potential of 500 volts direct current (d.c.).

3.4.7 Operating life. - The firing switch shall have an operating life of not less than 100 complete cycles as defined in 4.6.8.1.

3.5 Environmental requirements. - The firing switch shall meet the other applicable requirements of this specification after or while being subjected to environmental conditions defined below, when tested as specified in section 4.

3.5.1 Vibration. -

3.5.1.1 Vibration, transportation, storage and handling. - The firing switch, packed and packaged as defined in 5., mounted rigidly on a vibration device as specified in 4.6.1.1, shall withstand the vibrations for the time and at the temperatures to which a non-operating, ready non-operating or ready torpedo is exposed as specified in MIL-T-18404(NOrd) and thereafter shall show no damage and meet all the performance requirements and product characteristics of this specification.

3.5.1.2 Vibration, ready non-operating or ready condition. - The firing switch, less pack or packaging and rigidly self-mounted on a vibration device as specified in 4.6.1.2, shall withstand the vibrations for the time and at the temperatures to which a non-operating, ready non-operating or ready torpedo is exposed as specified in MIL-T-18404(NOrd) and thereafter shall show no damage and meet all performance requirements and product characteristics of this specification.

3.5.1.3 Vibration, operating condition. - The firing switch, self-mounted on a vibration device as specified in 4.6.1.3, shall withstand the vibrations of an operating torpedo for the time specified in MIL-T-18404(NOrd). These vibrations may occur in any direction. While undergoing and after the vibration test the firing switch shall

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show no damage and meet all performance requirements and product characteristics of this specification.

3.5.2 Shock

3.5.2.1 Shock, transportation, storage and handling. - The firing switch, packed and packaged as defined in 5., mounted rigidly on a shock testing device as specified in 4.6.2.1, shall be subjected to the shock requirements for a torpedo non-operating, ready non-operating or ready (transportation) as specified in MIL-T-18404(NOrd) and thereafter shall show no damage and meet all the performance requirements and product characteristics of this specification.

3.5.2.2 Shock, operating condition. - The firing switch, self-mounted on a shock testing device as specified in 4.6.2.2, shall be subjected to the shock requirements for a torpedo ready or operating, submerged tube launching, as specified in MIL-T-18404(NOrd) and thereafter shall show no damage and meet all the performance requirements and product characteristics of this specification.

3.5.3 Acceleration. - The firing switch, mounted securely on an acceleration device as specified in 4.6.3, shall be subjected to the acceleration requirements for a torpedo ready non-operating, ready or operating as specified in MIL-T-18404(NOrd). Thereafter the firing switch shall show no damage and meet all performance requirements and product characteristics of this specification.

3.5.4 Temperature and Humidity. -

3.5.4.1 Transportation and storage condition. - The firing switch, packaged and packed as defined in 5., shall be subjected to the temperature and humidity conditions specified in 4.6.4.1 and thereafter shall show no corrosion or damage and meet all the performance requirements and product characteristics of this specification.

3.5.4.2 High and low temperature operation. - The firing switch, less packaging and packing and mounted as specified in 4.6.4.2, shall be subjected to the high and low temperature conditions specified in 4.6.4.2. At the conclusion of the test, while still at the test temperature, the firing switch shall show no corrosion or damage and meet all the performance requirements and product characteristics of this specification.

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3.5.5 Pressure. -

3.5.5.1 Altitude. - The firing switch shall be subjected to the pressure requirements for torpedoes subject to air transportation as specified in MIL-T-18404(NOrd). Thereafter the firing switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.5.2 Snorkeling. - The firing switch shall be subjected to the pressure requirements for torpedoes subject to snorkeling submarine transportation as specified in MIL-T-18404(NOrd). Thereafter the firing switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.5.3 External air pressure. - The firing switch shall be subjected to the pressure requirements for torpedoes submarine launched as specified in MIL-T-18404(NOrd). Thereafter and while exposed to this pressure the firing switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.6 Tropical environment. - The firing switch shall be exposed to tropical environment conditions as specified in 4.6.6. Thereafter the firing switch shall show no fungus growth, corrosion or other damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.7 Marine atmosphere environment. - The firing switch shall be subjected to simulated sea atmosphere conditions and tests specified in 4.6.7 and thereafter show no evidence of corrosion or damage and meet the performance requirements and product characteristics of this specification.

3.5.8 Storage life. - The firing switch, packed and packaged in accordance with 5.1.1, shall be operable and meet all the performance requirements and product characteristics of this specification after being stored for a period of at least eight years under temperature and humidity conditions as follows:

3.5.8.1 Temperature: Minus 65 degrees F to plus 160 degrees F (minus 53.89 degrees C to plus 71.11 degrees C).

3.5.8.2 Humidity: Up to 95 per cent relative (maximum).

3.5.8.3 To meet the storage life requirement, rubber "O" rings shall be replaced after every five years of continuous storage.

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3.6 Clean-up. - Prior to and after final assembly, all components shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. - The firing switch, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, staking, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection, shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of five (5) firing switches shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for preproduction tests at an activity designated in the contract or by the Bureau of Ordnance, for purposes of determining compliance with the requirements of the contract, specifications and drawings. Further production of the firing switch by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. - As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot"; i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspec-

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tion lots" will be as determined by the Government inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.4 Periodic production samples. - From each succeeding lot or grand lot of three hundred (300) or less, a random sample of five (5) firing switches will be selected by the Government inspector for submission to a Bureau of Ordnance assigned activity to determine compliance of the samples with the requirements of the contract, specification, and drawings.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of the firing switch shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on firing switches being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Acceptance tests shall either be performed by the manufacturer and witnessed by the Government inspector, or shall be conducted by the Government inspector. These tests are defined in 4.5. Failure in any test will cause rejection of the firing switch.

4.3.2 Preproduction and periodic production tests. - Preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4 which are representative of the production of the firing switch after the award of the contract to determine that the lot and production meet the requirements of this specification. These are detailed in 4.6. Failure of any sample unit in any requirement or test shall result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.1 A suitable torque indicating instrument capable of measuring torques in the order of 2.5 inch pounds.

4.4.2 A device for exerting a pull of 1 pound on a wire in a direction along the axis of the wire.

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4.4.3 An evacuation chamber that allows observation of its interior, together with a vacuum pump and suitable gauge, capable of registering reduced pressures in the order of 15 inches of mercury.

4.4.4 A centrifuge disposed to rotate in a horizontal plane and capable of imparting accelerations up to 10 gravity units. Provision shall be made for mounting the firing switch in a socket with its axis parallel to the plane of rotation while its terminals are pointed toward the pivotal axis, and with its axis perpendicular to the plane of rotation. In the perpendicular mounting, additional provision shall be made for rotating the switch about its axis in seven other positions at increments of 45 $\pm$ 1 degree. Slip rings and brushes or some other appropriate arrangement shall be provided for making electrical contact with the rotating switch. The centrifuge itself shall be Government furnished equipment (GFE) similar or equal to the Genesco, Los Angeles, California Model B 78, 0-120 G.

4.4.5 An oscilloscope whose ripple is limited to less than 2.0 per cent. Associated with the oscilloscope is an electrical circuit containing a 1-1/2 volt d.c. power source rated at 1.5 amperes, the firing switch socket on the centrifuge, and a 1 ohm resistor, all connected in series. The d.c. potential drop across this 1 ohm resistor is applied to the vertical deflection plates of the scope. In addition a single pole, double throw, three-position "calibration" switch shall be provided in the circuit. When the calibration switch is in the "zero ohm" position, it shorts the firing switch socket to complete the circuit. When the calibration switch is placed in the "one ohm" position, the circuit is completed through a 1 ohm precision resistor connected in parallel with the firing switch socket. When the calibration switch is in the neutral or "operate" position, the circuit is completed by closure of the firing switch plugged into its socket. If resistances in the order of tenths of an ohm are introduced to the circuit through the use of slip rings, brushes, wiring, or by any other method, an additional means shall be provided for shorting the firing switch as close to its socket as is physically possible. OD 6474 shall be used as a reference document to establish equipment and circuitry considered satisfactory for the verification of switch performance.

4.4.6 A megohmmeter with an integral source of 500 volts d.c. and a mid-scale reading in the order of 10 megohms.

4.4.7 A standard electrical power source of 110 volts a.c., 60 cycle, for operation of the vacuum pump, centrifuge motor, oscilloscope, etc.

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4.5 Acceptance tests

4.5.1 Test conditions. - Unless otherwise specified, the firing switch shall be subjected to acceptance tests under the following conditions:

4.5.1.1	Temperature:	Room ambient plus 65 degrees F to plus 95 degrees F (plus 18.33 degrees C to plus 35.0 degrees C.)
4.5.1.2	Altitude:	Normal ground
4.5.1.3	Vibration:	None
4,5.1.4	Humidity:	Room ambient to 95 per cent relative (maximum)

4.5.2 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment and facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, where definite requirements are not specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.3 Test procedure. - The acceptance tests shall be performed in accordance with the following procedures:

4.5.3.1 Ball staking test. - This test shall be performed on the weight assembly, BUORD Drawing 1263258, before it is hermetically sealed inside the shell. With the use of a suitable torque indicating instrument (4.4.1), apply a 2.5 inch pound torque to the ball, first in one direction and then the opposite direction about the axis of the switch. When applying the torque, the ball shall not be gripped in such a manner as to scratch, mar or in any way damage the gold plated surface.

4.5.3.2 Pigtail wire securement test. - This test shall also be performed on the weight assembly, BUORD Drawing 1263258, before it is hermetically sealed inside the shell. Apply a pull of 1 pound (4.4.2) to the pigtail wire along the axis of the weight base.

4.5.3.3 Airtightness test. - This test shall be performed upon the firing switch assembly, BUORD Drawing 1390702, before it is finally assembled into the plastic case. Fill a container with water to sufficient depth so that the assembly may be completely submerged. Place this water bath in the evacuation chamber (4.4.3) and reduce the air

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pressure to at least 15 inches of mercury in order to remove the air which is dissolved in the bath. Bring the pressure back to ambient and submerge the firing switch assembly in the deaerated water bath. Evacuate the chamber to a pressure of 15 inches of mercury (maximum), and observe for a period of 2 minutes. Leakage from the switch assembly will be shown by escaping air bubbles.

4.5.3.4 Sensitivity and contact resistance test. -

4.5.3.4.1 Calibration of oscilloscope. - First it is necessary to calibrate the oscilloscope (4.4.5) to represent a 1 ohm deflection and fractions thereof. In the electrical circuit (4.4.5) associated with the oscilloscope, throw the calibration switch to the "zero ohm" position, and adjust the scope so that the d.c. presentation appears on the lower portion of the screen. Then throw the calibration switch to the "one ohm" position and adjust the scope amplification so that the d.c. presentation appears on the upper portion of the screen. Continue adjusting the magnitude and position of the scope deflection until repeated operation of the calibration switch separates the d.c. presentation by ten units according to some convenient scale.

4.5.3.4.2 Conversion of deflection to resistance. - When the calibration switch is in the "zero ohm" position, the total resistance of the calibration circuit is 1 ohm plus the resistance of the wiring. When the calibration switch is in the "one ohm" position, the total resistance of the calibration circuit is greater by exactly 1 ohm. The increase in resistance is represented by a scope deflection of ten units. Compute the number of units of scope deflection corresponding to each 0.1 ohm increase in the calibration circuit resistance between the "zero ohm" position and the "one ohm" position. Plot the results on a graph showing resistances from 0 to 1 ohm versus scope deflection. Thus, an easy conversion may be made from units of scope deflection to resistances of a fraction of an ohm.

4.5.3.4.3 Conversion of revolutions per minute (RPM) to gravity units. - Draw up another table or graph of RPM versus gravity units according to the formula

$$A = 3.4 \times 10^{-4} RN^2$$

where the following units apply:

A = gravity units of acceleration
R = radial arm of centrifuge in feet; the radial arm of the centrifuge shall be defined as the perpendicular distance from the axis of rotation to the

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center of gravity of the moving parts of the switch when the moving parts are at rest.

N = revolutions per minute

If the centrifuge does not register RPM directly, but indicates angular velocity in some arbitrary unit, it will be necessary to obtain a correlation between these units and the RPM.

4.5.3.4.4 Test. - Mount the firing switch in the socket on the centrifuge (4.4.4) so that its axis is perpendicular to the plane of rotation. Throw the calibration switch into the neutral or "operate" position and start the centrifuge. After attaining an acceleration of 2 gravity units, increase the acceleration thereafter at a rate not exceeding 0.1 gravity unit per second. When a pip is observed on the oscilloscope screen, note the reading on the centrifuge indicator. Convert this reading to acceleration with the aid of the graph prepared in 4.5.3.4.3 to obtain the value for instantaneous closure. Slowly increase the acceleration within the specified rate. When a straight line d.c. presentation is observed on the oscilloscope screen, make a notation of the number of units of scope deflection and again note the reading on the centrifuge indicator. Convert these readings to resistance and acceleration with the aid of the graphs prepared in 4.5.3.4.2 and 4.5.3.4.3. The acceleration obtained is the value for steady closure and the resistance value obtained is a combination of switch contact resistance plus resistance of the circuit. If the particular circuitry warrants it, short out the firing switch as close to the socket as is physically possible, and obtain a scope deflection for the circuit resistance alone while the centrifuge is still at the same speed. Convert this deflection to resistance and subtract this resistance from the value previously obtained to determine true switch contact resistance in the position tested. While perpendicularly mounted, test the firing switch for instantaneous and steady closure acceleration and contact resistance in seven other positions at 45 $\pm$ 1 degree increments of rotation about its axis. Check the calibration of the scope deflection before every test. Then mount the switch so that its axis is parallel to the plane of rotation and the switch terminals toward the pivotal axis of the centrifuge. Conduct one test in this position.

4.5.3.5 Dielectric test. - Apply the test probes of a megohmmeter (4.4.6) to the terminals of the firing switch when it is in the normally open condition.

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4.6 Preproduction and periodic production tests.

4.6.1 Vibration tests. -

4.6.1.1 Vibration, transportation, storage and handling. - The firing switch, inside of its package, with the package mounted rigidly on the vibration table to simulate as realistically as possible the service conditions, shall be subjected to vibration testing for the time periods and at the temperatures specified in MIL-T-18404(NOrd) for a torpedo non-operating, ready non-operating or ready.

4.6.1.2 Vibration, ready non-operating, or ready condition. - The firing switch, mounted rigidly on the vibration table to simulate as realistically as possible the service installation, shall be subjected to vibration testing for the time periods and at the temperatures specified in MIL-T-18404(NOrd) for a torpedo non-operating, ready non-operating or ready.

4.6.1.3 Vibration, operating condition. - The firing switch, mounted rigidly on the vibration table to simulate as realistically as possible the service installation, shall be subjected to vibration testing for the time periods specified in MIL-T-18404(NOrd) for a torpedo operating.

4.6.2 Shock tests. -

4.6.2.1 Shock, transportation, storage and handling. - The firing switch, inside of its package, with the package mounted rigidly on the shock testing device to simulate as realistically as possible the service conditions, shall be subjected to transportation shock as specified in MIL-T-18404(NOrd).

4.6.2.2 Shock, operating condition. - The firing switch, mounted rigidly on the shock testing device to simulate as realistically as possible the service installation, shall be subjected to the water-entry shock for a deck launched torpedo as specified in MIL-T-18404(NOrd) for a torpedo non-operating, ready non-operating, ready or operating.

4.6.3 Acceleration test. - The firing switch, mounted on the testing device to simulate as realistically as possible the service installation, shall be subjected to acceleration testing as specified in MIL-T-18404(NOrd) for a torpedo ready non-operating, ready, or operating.

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4.6.4 Temperature and humidity tests. -

4.6.4.1 Transportation and storage temperature test. - The firing switch, when inside of its package, shall be subjected to temperature and humidity cycling as specified for a non-operating torpedo in MIL-T-18404(NOrd).

4.6.4.2 High and low temperature tests. - The firing switch, mounted in a simulated well or switch cavity to simulate as realistically as possible the service installation, shall be subjected to low temperature testing as specified in MIL-T-18404(NOrd) for a ready or operating torpedo, submarine launched, and to high temperature testing as specified for a ready non-operating, ready, or operating torpedo.

4.6.5 Pressure tests. -

4.6.5.1 Altitude test. - The firing switch shall be subjected to reduced air pressure testing as specified for torpedoes subject to air transportation in MIL-T-18404(NOrd).

4.6.5.2 Snorkeling test. - The firing switch shall be subjected to pressure testing as specified for torpedoes subject to snorkeling submarine transportation in MIL-T-18404(NOrd).

4.6.5.3 External air pressure test. - The firing switch shall be subjected to external air pressure testing as specified in MIL-T-18404(NOrd) for torpedoes that are submarine launched.

4.6.6 Tropical environment test. - The firing switch shall be exposed to fungus as specified in MIL-T-18404(NOrd) for a torpedo non-operating or ready non-operating.

4.6.7 Marine Environment Test. - The firing switch shall be subjected to the simulated sea atmosphere tests under conditions specified in MIL-T-18404(NOrd).

4.6.8 Operating life. -

4.6.8.1 Definition of one cycle. - One cycle of operation for the firing switch shall be defined as the discharging of a 4.0 microfarad (MFD) capacitor having an initial charge of 150 volts d.c. through a 5 ohm resistor with one impact applied to the switch.

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4.6.8.2 Operating life test. - Mount the firing switch in a test fixture that will impart sufficient shock to trip the switch. Connect the circuit (4.6.8.1) for discharging the 4.0 MFD capacitor upon switch closure. Subject the firing switch to 100 cycles of operation, rotating the switch approximately 3-1/2 degrees with respect to the line of impact prior to each successive cycle. The minimum time between each impact shall be 1 minute.

4.6.9 Workmanship test. - Secure the assembled switch in a lathe, or like tool, and cut in half at the 0.062 radius spherical dimple. While carefully separating the halves, observe for the presence of dirt, loose or excessive solder, flux, and loose or frayed strands of pigtail wire. This test shall be conducted in accordance with MIL-STD-105, Table VI-C, AQL 4.0 per cent defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Level A - Unless otherwise specified, the firing switch shall be packaged and packed as an integral component of the exploder mechanism in the containers specified for the exploder mechanism as set forth in the applicable BUORD drawings.

5.1.2 Level B - Not applicable

5.1.3 Level C - Not applicable

5.1.4 Spare or replacement parts. - When procured as spare or replacement parts, firing switches shall be packaged and packed as specified by the procuring activity.

5.2 Labeling and marking. - Unless otherwise specified, labeling and marking of all shipping containers shall be in accordance with MIL-STD-129. Each container shall be marked with the serial number of the unit it contains.

6. NOTES

6.1 Intended use. - The Firing Switch Mark 3 Mod 6 is intended for use in the Exploder Mechanism Mark 11 series.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.

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(b) Any exceptions to this specification, applicable drawings or other documents.

(c) Applicable preservation, packaging and packing if firing switches are procured as spare or replacement parts. See 5.1.4

(d) The centrifuge required to test these switches shall be Government furnished equipment. See 4.4.4

NOTICE:

When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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